

Forge

*Manufacturing a private, on-device PII redaction specialist —
and what measuring it honestly revealed about its own teacher.*

1.000

MIN HIGH-SEVERITY RECALL
(SYSTEM)

0.9482

TEACHER MICRO-F1

574

FROZEN GOLD RECORDS

Technical Report

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Task: on-device PII detection & redaction, 19 entity types

Teacher: openai/gpt-oss-120b (Apache-2.0) · Student: Qwen2.5-1.5B-Instruct + LoRA

1. Executive summary

Forge is a task-specialization distillation pipeline. It takes one expensive, high-volume LLM task and an open teacher model, and manufactures a small specialist that runs fully offline. The flagship task is on-device PII detection and redaction, where the privacy argument is structural: you cannot send sensitive text to a frontier API *in order to find the sensitive text*.

The project's most valuable output was not the model. It was a **negative finding about the teacher**, and the engineering response to it.

The finding

The teacher scores a strong **0.9482 micro-F1** overall, but misses **6 of 9 breach-severity recall floors** — types where a single miss is a reportable disclosure. Driver's licence recall is **0.5333**.

Because distillation transfers a teacher's blind spots, **no student trained on this teacher could ever clear those floors**. A student at *perfect* parity would still inherit 0.53 recall on driver's licences. The gate was unreachable by distillation, and moving it was not permitted — this project voids any run whose threshold is renegotiated after seeing results.

Response: stop asking a language model to do arithmetic. Deterministic validators (Verhoeff, Luhn, format + nearest-keyword context) carry the nine high-severity types; the distilled model keeps the contextual ones. All nine floors reach **1.0000**.

This outcome was predicted before it was measured. The project's honest-assessment document, written earlier, stated: “for well-formed identifiers, a well-written regex with a checksum is likely to beat a 1.5B model.” The data agreed.

2. Measured results

Frozen 385-record test set. Exact-match (start, end, label) micro-F1. Identical harness for every row. The teacher was scored *before* the student finished training, so the parity threshold could not be back-fitted.

System	micro-F1	precision	recall	min high-sev recall
Teacher — gpt-oss-120b	0.9482	0.9615	0.9353	0.5333
Student, model only	0.5750	0.6375	0.5237	0.0000
Forge system (student + validators)	0.7334	0.7818	0.6906	1.0000

Three numbers are always published together — model-only, validator-only, and system. Quoting the system score as though it measured the distillation would be exactly the conflation this project's gate discipline exists to prevent. **G1 parity is measured on model-only, and model-only is 0.5750 against a 0.9292 target: the parity gate is not met.**

Breach-severity types (contract floor: 0.99 recall)

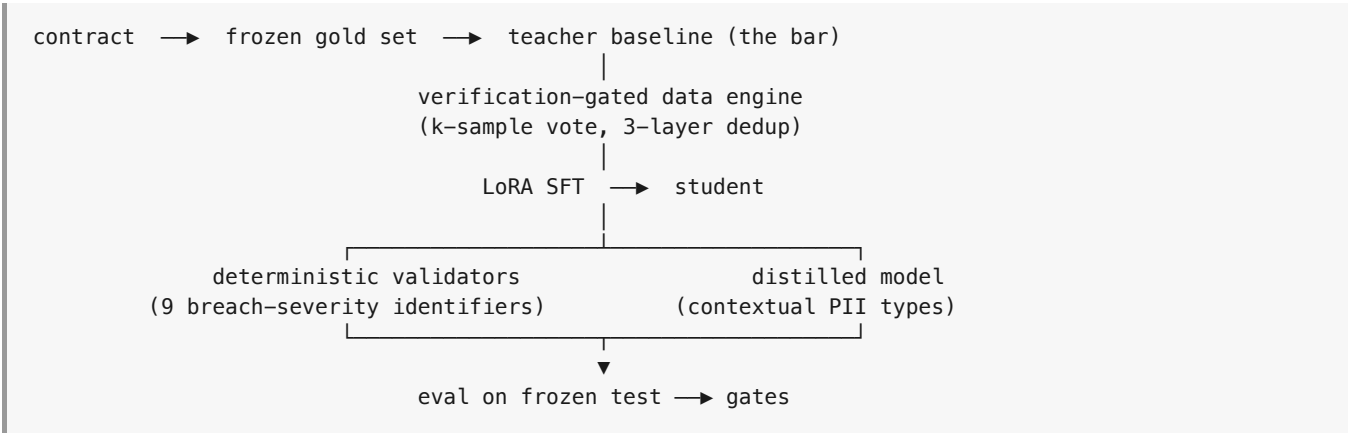
Type	Teacher	Student only	Forge system
DRIVER_LICENSE	0.5333	0.0000	1.0000
BANK_ACCOUNT	0.7931	0.2414	1.0000
AADHAAR	0.8276	0.1724	1.0000
PASSPORT	0.9130	0.6957	1.0000
PAN	0.9310	0.5862	1.0000
PASSWORD	0.9474	0.2632	1.0000
CREDIT_CARD	1.0000	0.6341	1.0000
SSN	1.0000	0.6111	1.0000
API_KEY	1.0000	0.4483	1.0000
minimum	0.5333	0.0000	1.0000

The student alone found **zero** of fifteen driver's licences. Without the validator layer this system would leak breach-severity identifiers at scale, which makes that design decision load-bearing rather than an optimisation.

Gate status

Gate	Threshold	Measured	Verdict
High-severity recall	≥ 0.99 on 9 types	1.0000	PASS
G2 schema validity	≥ 99.9%	384/385	marginal
G1 quality parity	≥ 0.9292	0.5750	OPEN
G4 latency (p95)	≤ teacher/5	teacher 8.02s	pending
G3 cost, G5 deploy, G6 OOD	—	harnesses built	pending

3. Architecture



The ordering is the design. Scoring the teacher first means the parity threshold is a consequence of measurement rather than a target chosen to be reachable.

Why checksums are a precision signal, never a recall gate

Only **2 of 29** Aadhaar values in the gold set satisfy the Verhoeff checksum — the synthetic generator emits random digits, while every *real* Aadhaar number is checksummed. Gating detection on the checksum would score 0.07 recall here and near-1.0 on real data: a validator whose recall silently depends on which dataset it meets. So a failing checksum lowers confidence and is reported; it never suppresses a span.

Where the checksum does earn its keep is *disambiguation*. Both 12-digit numbers the gold set labels as cards are Luhn-valid, and the 14-digit number it labels a bank account is not — which resolves the collision correctly without suppressing anything.

4. What the distillation experiments established

Two training runs, each with predictions registered *before* execution, each rejecting its own hypothesis.

Run	Change	Train loss	F1	Verdict
run_002	5.6× targeted data (150 → 837)	1.17 plateau	0.5750	underfit — data was not the constraint
run_003	17× adapter capacity (r=16→64, +MLP)	0.22	0.5379	overfit — recall collapsed

run_003's loss fell roughly 5× while F1 got *worse*: precision +0.24, recall −0.13, and the span ratio halved from 0.84 to 0.46. The governing ADR had written the interpretation in advance — “fitting the data better without learning to enumerate entities” — so the result was diagnostic rather than confusing.

Why the conjunction mattered

That ADR required *both* a loss improvement and a span-ratio improvement. Had it asked only “does loss improve?”, run_003 would have read as a clean success and the recall collapse would have shipped undetected into the next round.

Together the runs show capacity and data are **jointly binding**: adding capacity alone trades recall for precision at roughly constant F1, which is movement along a frontier rather than progress toward it. That isolates the remaining lever, and it is one the project has never pulled — every training record is template-generated, and the 150 “teacher-labelled” ones came from an 8B development model later replaced for being too weak. **No training record has been labelled by the 0.9482-scoring teacher.** A project named for distillation has not yet distilled from its teacher.

5. Engineering practices

A reproducibility defect, found and fixed

The “frozen” gold set was silently drifting. Faker's `date_of_birth()` derives its sampling window from `datetime.now()`, so the seed fixed *where in the window* a value fell while the window itself slid one day per

day. The set was reproducible *within* a day and different across days — measured as an exact +4-day skew four days after the build, and +8 after eight.

This falsified the contract's core promise and would have handed any stranger cloning the repo a different test set. The fix anchors to a fixed epoch while keeping Faker's own call path, so the committed data reproduces **bit-for-bit** and no prior measurement was invalidated. A clock-shifted regression test, verified to fail against the old code, prevents recurrence.

Discipline that is enforced, not asserted

- **Gates are never moved.** When the teacher changed, contract v2 superseded v1 rather than editing it, with all six thresholds verified byte-identical programmatically.
- **Honest instrumentation.** The AWQ export path refuses to run rather than silently skipping when CUDA is absent. The economics harness prices the teacher at *paid* rates despite development running on a free tier, because a subsidy is not an economics claim.
- **Failure-tolerant pipelines.** Every long stage resumes; machine sleep destroyed three separate multi-hour runs before per-record flushing and step checkpointing were added.
- **12 architecture decision records**, including two that record the rejection of their own hypothesis.

6. Honest assessment

Roughly 85% of this work is standard practice: distillation, LoRA, self-consistency filtering, active learning, n-gram dedup. Using a library is not a contribution. What is less common is the *enforcement* — pre-committed gates that require a version bump to change, construction-verified data chosen because the teacher was measurably weak on the needed types, and economics treated as a gate with published arithmetic.

Known weaknesses

- **The parity gate is not met.** 0.5750 against 0.9292. The pipeline is proven to run; it is not yet proven to reach parity.
- **Evaluation is synthetic.** Faker values in templates — reproducible and leak-free, but not natural text, so scores overstate real-world performance.
- **The gold set has had no documented human verification pass.** Until it does, “human-verified” is not a claim this project may make.
- **Synthetic identifiers are not structurally valid** (2/29 Aadhaar pass Verhoeff), so validator precision on real input would likely exceed what is reported here.
- **Single task, single language.** Nothing here demonstrates the pipeline generalizes to a second task.
- **QLoRA and AWQ are unrun** — both require CUDA, unavailable on the development hardware. Blocked, not pending.

The uncomfortable comparison

For well-formed identifiers, a checksummed regex likely beats a 1.5B model on precision, latency, and cost simultaneously. This project's own results support that, which is why the validator layer exists. A defensible version must eventually show the model winning on *contextual* PII where rules fail. That has not been measured, so “specialist model beats the alternatives” is not a claim this project has earned — only “specialist system carries the safety-critical floor” is.

7. Reproducing

```
git clone https://github.com/Jatin-IITB/forged && cd forge
make install
export CEREBRAS_API_KEY=...      # free tier
make forge
```

Every stage resumes. The teacher endpoint is fungible — the same Apache-2.0 checkpoint is served by multiple providers and self-hostable with vLLM, so `TEACHER_URL` can point anywhere. **Independence litmus test:** if every private credential were revoked tomorrow, a stranger could still clone this repo and rebuild it end to end.

8. Decision records

Every non-obvious choice is recorded with its context, the alternatives considered, and the consequences accepted. Two entries document the rejection of their own hypothesis.

ADR	Decision
0001	Eval-first: the gold set and parity gate exist before the model
0002	Verification gate on all training data
0003	Zero internal dependency: public data + open weights + own compute
0005	Flagship task, teacher/base models, and data source
0006	Patterns adopted from public prior art
0007	Rationale-augmented distillation
0008	Training pipeline design (SFT + LoRA)
0009	Error-driven active learning (Phase 4)
0010	Teacher scale-up: Qwen3-8B → GPT-OSS-120B (Cerebras free tier)
0011	The frozen gold set was not actually frozen (clock-dependent generation)
0012	High-severity identifiers move to a deterministic validator layer
0013	run_002 diagnosis: the bottleneck is adapter capacity, not training data